DAQ 2025s

MechPulse

Multi-Sensor Machine Health Monitor

Team 22 : GK Fastfood

Kawin KAEWPARADAI · Kantee LAIBUDDEE

KY-002 · KY-028 · KY-037 · KY-031 → ESP32 → MQTT → Node-RED → MySQL → FastAPI → React

2.1 Overview & Motivation

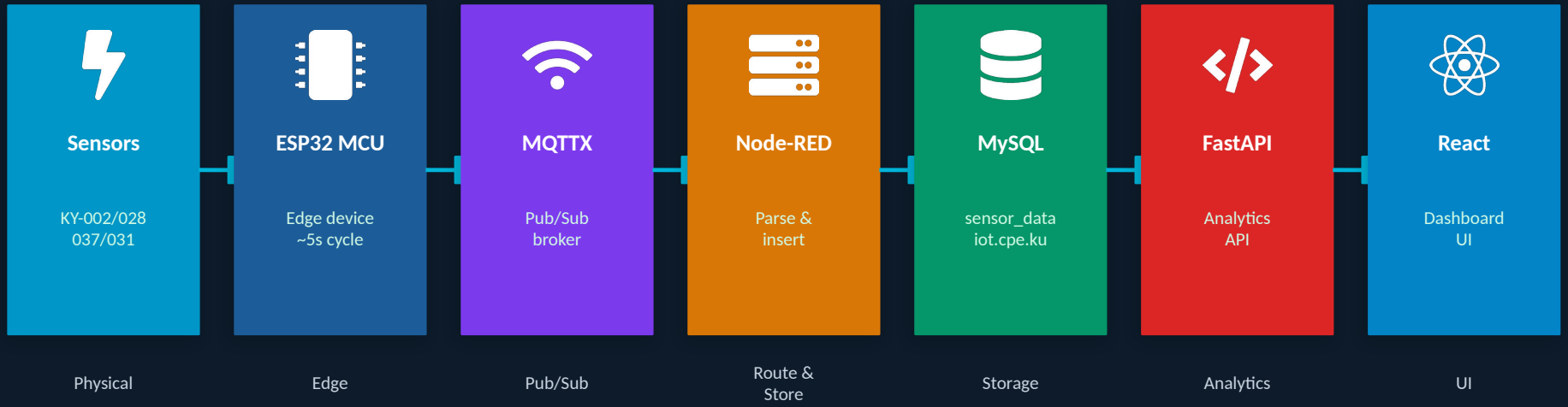
The Problem

- Rotating machinery (motors, fans) fails without warning.
- Unexpected downtime causes costly damage and production loss.
- Manual inspection is slow and misses early-stage faults.
- No single signal is sufficient — combined sensing gives higher accuracy.

MechPulse Solution

- 4 KY sensors on ESP32 sample every ~5 seconds.
- Node-RED routes data via MQTT into a shared MySQL DB.
- FastAPI computes a 0–100 Health Score and z-score anomalies.
- React dashboard gives real-time visibility: time-series, heatmap, alerts, daily reports.

2.2 Overall Architecture



Each layer has a single responsibility — any component can be swapped independently.

2.3 Primary Data Source — KY Sensor Suite

KY-002 · Vibration

Analog vibration amplitude. Measures intensity & frequency patterns of rotating parts.

KY-028 · Temperature

Digital surface temperature. Detects overheating above reference threshold (45 °C).

KY-037 · Sound Level

Analog sound level. Captures grinding, rattling and abnormal acoustic signatures.

KY-031 · Knock/Impact

Digital tap detection. Flags sudden mechanical shocks or irregular impacts.

Collection Method: ESP32 reads all 4 sensors on a fixed ~5 s cadence → assembles JSON payload → publishes to MQTT topic mechpulse/<machine_id>/reading

2.3 Secondary Data Sources

Manufacturer Specifications

Reference operating values stored as REF_* environment variables:

Variable	Value	Meaning
REF_TEMP_C	45	Max normal temp (°C)
REF_VIB_COUNT	5	Max vibration count
REF_SOUND_AVG	870	Normal sound avg
REF_SOUND_VAR	60	Normal sound variance

Historical Baseline Phase

The first N rows where phase = "baseline" define calibration statistics:

- **Mean (μ) and Standard Deviation (σ)**
- Used by the /api/analytics/anomalies endpoint to compute per-column z-scores.

Z-score formula:

$$z = (x - \mu) / \sigma$$

- Severity levels: info → warning → critical
- Cross-validation: severity upgraded when ≥ 2 sensors fire at the same timestamp.

2.4 Database Schema — sensor_data

MySQL — iot.cpe.ku.ac.th/pma

Table: **sensor_data**
Index: idx_sensor_ts (timestamp)

Column	Type	Description
id	INT AUTO_INCREMENT PK	Unique row identifier
timestamp	DATETIME NOT NULL	Indexed — time of reading
phase	VARCHAR(20)	baseline / warmup / running / cooldown
cycle	INT	Monotonic counter within phase
vib_count	INT	Vibration event count (KY-002)
knock_count	INT	Knock/impact count (KY-031)
sound_avg	INT	Average sound level (KY-037)
sound_var	INT	Sound variance (KY-037)
temperature_c	FLOAT NULL	Surface temperature °C (KY-028)

2.5 Data Sharing API — FastAPI + Node-RED

FastAPI Endpoints (OpenAPI docs at /docs)

Method	Path	Description
GET	/api/health	Liveness + DB connection check + row count
GET	/api/schema	Columns in the sensor_data table
GET	/api/sensors/latest	Most recent N rows
GET	/api/sensors/timeseries	Single-metric time-series (phase filter)
POST	/api/sensors/ingest	Insert a new row (MQTT bridge / testing)
GET	/api/analytics/stats	Count / mean / std / quantiles per column
GET	/api/analytics/correlation	Pearson / Spearman / Kendall matrix
GET	/api/analytics/health-score	Composite 0–100 health score + components
GET	/api/analytics/anomalies	Z-score anomalies: info / warning / critical
GET	/api/analytics/daily-summary	Peak vib, avg sound, impact count per day

Node-RED also exposes HTTP JSON endpoints: GET /api/sensor and GET /api/weather (both with optional ?limit=N)

2.6 Data Visualization — React Dashboard

Dashboard

- Live health score (0-100)
- Recent time-series charts
- Anomaly alert panel
- Latest sensor stats

Analytics

- Pearson/Spearman/Kendall heatmap
- Descriptive stats table
- Column correlation explorer
- Filter by phase

Architecture

- Interactive SVG pipeline diagram
- Layer descriptions
- Component responsibilities
- Stack documentation

Reports

- Daily summary table
- Peak vibration per day
- Anomaly log (info/warn/crit)
- Impact & sound trends

Stack: React + Vite + Tailwind CSS + Recharts + React Router · API polling every 10 s

2.7 Demonstration of Key Features

Machine Health Score (0–100)

Composite score from 4 weighted penalties: vibration excess, knock excess, sound deviation, temperature breach.
Score 100 = perfectly healthy. Drops when any metric exceeds its REF_* reference threshold.

92

Z-Score Anomaly Detection

Per-column z-scores vs baseline (mean + std). Severity: Info (z>2), Warning (z>3), Critical (z>4).
Cross-sensor validation: severity upgraded when ≥2 sensors fire at the same timestamp — reduces false positives.

Info

Warn

Crit

Correlation Heatmap — 3 Methods

Pearson, Spearman, and Kendall correlation matrices computed on demand.
Constant columns are auto-ignored. Renders as an interactive color heatmap in the Analytics page.

Thank You!

MechPulse — Multi-Sensor Machine Health Monitor

- 4 KY sensors · ESP32 · MQTT · Node-RED
- MySQL (iot.cpe.ku.ac.th) · FastAPI · React + Recharts
- Health Score · Z-Score Anomalies · Correlation Matrix
- Source Code: GitHub (Tag: v1.0) · DB: sensor_data table